Extended Kalman Filter Implementation

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Definitions

Estimated State Vector

$$\hat{x}_k = \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix}$$

The estimated state vector is composed of the estimated position, velocity, and acceleration at time k in a column matrix.

Process Noise w

$$w_k = \begin{bmatrix} w \\ \dot{w} \\ \ddot{w} \end{bmatrix}$$

w_k denotes the process noise. w, w dot, and w double dot denote noise in position, velocity and acceleration respectively. In our implementation, only acceleration noise matters. None of these values need to be found, this matrix only exists to derive Jocobian W discussed later.

Input u

$$u_k$$

Input u_k is a scalar. It reports the acceleration measured by the IMU.

Projection Function f

$$f(\begin{bmatrix} x_{k-1} \\ \dot{x_{k-1}} \\ \ddot{x_{k-1}} \end{bmatrix}, u_{k-1}, w_{k-1}) = \begin{bmatrix} x_{k-1} + t\dot{x}_{k-1} + \frac{1}{2}t^2\ddot{x}_{k-1} \\ \dot{x} + t\ddot{x}_{k-1} \\ \ddot{x} + u_{k-1} + \ddot{w}_{k-1} \end{bmatrix}$$

The projection function projects the previous state to the new state. w_{k-1} is also a column matrix but is not shown here. Small ‘t’ will be used throughout this document to represent the time between measurements in other words the ‘loop_time’.

This specific projection function was arbitrary. Other more sophisticated projections may take factors like drag into effect. It is very close to the original Kalman Filter function being used earlier. Switching to a more complicated projection function should be considered especially for rotation.

Jacobian A

$$A = \nabla f_x = \begin{bmatrix} 1 & t & \frac{1}{2}t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$$

A is the Jacobian of function f with respect to matrix x hat.

Jacobian W

$$W = \nabla f_w = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

W is the Jacobian of function f with respect to noise matrix w. This can be physically understood to mean that only the noise in acceleration, i.e. w double dot, affects our system.

Process Noise Q

Scalar matrix of the variance of the IMU. The value is experimentally determined and assigned to the matrix beforehand. This value does not change during the running of the filter.

Had the projection function been different, each element of the diagonals would have been assigned the variance of its respective component, i.e. variance in position, velocity, and acceleration. As f only uses u in for acceleration, Jacobian W masks out diagonals 1,1 and 2,2 anyway. Hence, only the value of variance in acceleration, element 3,3 is important.

Error Covariance Matrix P

3x3 Matrix. Keeps changing while the filter is running. Calculated at both projection and correction.

Measurement z

$$z_k = \begin{bmatrix} x_{gps} \\ \dot{x}_{gps} \\ \dot{x}_{gps} \end{bmatrix}$$

Contained the state as measured by the GPS.

Measurement Noise v

$$v_k = \begin{bmatrix} v \\ \dot{v} \\ \ddot{v} \end{bmatrix}$$

Much like the process noise, the values for this vector do not need to be found. It only exists to be used to calculate a Jacobian.

Function h

$$h(\hat{x}_k^-, v_k) = \hat{x}_k^- + v_k = \begin{bmatrix} x_k + v \\ \dot{x}_k + \dot{v} \\ \ddot{x}_k + \ddot{v} \end{bmatrix}$$

Converts projected state to the same format as the measurement. In our model, they are already in the same format. Since v is assumed to be 0, wherever h(x,0) has to be used, the priori state can be used directly.

Jacobian H

$$H = \nabla h_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Jacobian of function h with respect to state vector x. Happens to be Identity Matrix for the chosen function h.

Jacobian V

$$V = \nabla h_v = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Jacobian of function h with respect to noise vector v. Happens to be Identity Matrix for the chosen function h.

Measurement Noise R

$$R = \begin{bmatrix} r_x & 0 & 0 \\ 0 & r_{\dot{x}} & 0 \\ 0 & 0 & r_{\ddot{x}} \end{bmatrix}$$

Diagonal Matrix of variance of measurement provided by GPS for position, velocity, and acceleration respectively. Currently only positional variance is found and then multiplied by looptime and looptime squared respectively to get the other two.

In the future experimentally determining the values of variance of velocity and acceleration is suggested.

Projection

Projecting State Ahead:

$$\hat{x}_k^- = f(x_{k-1}, u_{k-1}, 0)$$

Noise w_k is assumed to be zero for the projection.

In code:

```
  priori_PoseStateX(0) = PoseStateX(0) + PoseStateX(1)*loop_time +
(PoseStateX(2)*loop_time*loop_time)/2;
  priori_PoseStateX(1) = PoseStateX(1) + PoseStateX(2)*loop_time;
  priori_PoseStateX(2) = PoseStateX(2) + acc_X_IMU;
```

Projecting Error Covariance Ahead:

$$P_k^- = AP_{k-1}A^T + WQ_{k-1}W^T$$

In code:

```
PrioriErrorCov_P_X = Jacobian_A * ErrorCov_P_X * (~Jacobian_A) +
Jacobian_W * ProcessNoise_Q * (~Jacobian_W);
```

Correction

Computing Kalman Gain:

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + V_k R V_K^T)^{-1}$$

Substituting Identity matrix for H and V:

$$K_k = P_k^- (P_k^- + R)^{-1}$$

In code:

```
BLA::Matrix<3, 3> KalmanGain_X = PrioriErrorCov_P_X *
(invertMatrix(PrioriErrorCov_P_X + MeasNoise_R_Gps3));
```

Updating State with Measurement:

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - h(\hat{x}_k^-, 0))$$

Since h(x,0) is the same as x for our defined h(x,v) we can rewrite this as:

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - \hat{x}_k^-)$$

In code:

```
PoseStateX = priori_PoseStateX + KalmanGain_X * (measurementX -
priori_PoseStateX);
```

Updating Error Covariance:

$$P_k = (I - K_k H_k) P_k^-$$

H_k is removed as it is equal to I.
In code:

```
ErrorCov_P_X = (Identity - KalmanGain_X) * PrioriErrorCov_P_X;
```

References

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